

## Multiple Choice (10 marks)

1. Let `x = 8`. What is `x >> 1` in Python 3?
  - (a) 3
  - (b) 4
  - (c) 2
  - (d) 8
2. Which of the following best explains what happens when a new device is connected to the Internet?
  - (a) A device driver is assigned to the device.
  - (b) An Internet Protocol (IP) address is assigned to the device.
  - (c) A packet number is assigned to the device.
  - (d) A Web site is assigned to the device.
3. What is the output of `"abc"[-1]` in Python 3?
  - (a) Error
  - (b) a
  - (c) b
  - (d) c
4. In Python 3, which of the following function removes all leading and trailing whitespace in string?
  - (a) `replace(old, new [, max])`
  - (b) `strip([chars])`
  - (c) `swapcase()`
  - (d) `title()`
5. Which of the following best describes the primary way in which a distributed denial-ofservice (D D o S) attack differs from a denial-ofservice (D o S) attack?
  - (a) The goal of the attack
  - (b) The number of computers being attacked
  - (c) The number of computers launching the attack
  - (d) The time period in which the attack occurs

## True / False (10 marks)

*Answer True or False*

6. True or False: In Python 3, the expression `['Hi!'] * 4` produces an error due to incorrect syntax.
7. True or False:  $O(1)$  represents constant time complexity, making it the smallest asymptotic notation among common complexities.
8. True or False:  $O(\log n)$  is larger than  $O(n^2)$  as  $n$  increases, indicating faster growth.
9. True or False: In Python 3, floor division is performed using the operator `'/'`.
10. True or False: A function that sorts values in a column or row can help identify unusually high or low data entries.

## Long Form Response (20 marks)

*Answer each question with a clear and complete explanation.*

11. Given the list `l = [1, 2, 2, 3, 4]` in Python 3, discuss how the set data structure processes this list and explain the significance of the output.
12. Discuss how the trend of computer processing speeds doubling every two years can influence technology companies' research and development strategies. Provide examples to support your analysis.

*End of Paper*